Simon Xia

Design Engineering Undergraduate

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Design Engineering undergraduate at Imperial College London seeking an industry placement in mechanical engineering and engineering analysis. Project experience spans surface modelling, computational fluid dynamics, sustainable product architecture and physical prototyping. Uses quantitative analysis and iterative evaluation to connect design decisions with functional requirements.

EDUCATION

Imperial College London 2024 - 2028 (expected)
Design Engineering | Four-year degree

Guanghua International School Shanghai, China

TECHNICAL SKILLS

CAD and visualisation	Fusion 360, Blender; surface modelling, assemblies and rendering
Engineering analysis	ANSYS Fluent, MATLAB; CFD, numerical analysis and data interpretation
Programming	Python, C++, JavaScript
Electronics and UI	KiCad for PCB design; Figma for interface design

SELECTED ENGINEERING PROJECTS

STREAM X
Vehicle aerodynamics | Individual

- Developed a continuous-surface vehicle concept with an integrated roof airflow channel, using morphological analysis to evaluate body-form and airflow-control alternatives.
- Configured a reduced-scale, symmetry-based ANSYS Fluent model at 30.5 m/s with k-epsilon turbulence modelling and a 458,336-cell polyhedral mesh.
- Analysed pressure and velocity fields to examine separation and rear-wake behaviour; documented model-scale limitations and proposed underbody and channel refinements.

MODUFRAME
Sustainable aircraft seating | Individual

- Defined a modular aircraft-seat architecture with an aluminium 6061 frame, replaceable comfort layers and reversible mechanical connections to support maintenance and material separation.
- Evaluated material and manufacturing impacts using Ansys Granta EduPack Eco Audit and a mass-weighted Material Circularity Indicator; examined trade-offs between mass, durability and material recovery.

AxisPro
Electromechanical product | Four-person team

- Contributed to a motorised miniature-painting workstation with height, tilt and rotation control, developed through team prototyping and ergonomic evaluation.
- Led technical rendering, product labelling and a five-page user guide covering setup, operation, safety and maintenance; developed the product business model.

SIMREST Q2
Product concept | Individual

- Developed a compact bedside-device enclosure through CAD iteration, comparing slider and push-button interfaces and communicating component arrangement with exploded and dimensioned views.

Jìng The Path
Spatial design | Five-person team

- Contributed to a campus-space proposal using CAD, scale models and full-size cardboard prototypes; the team used feedback from 12 co-design participants to refine seating and layout.